

# SCH4U DIY Thesis Lab: Pre-Lab

## A Non-Standard Vinegar Cell and the Nernst Equation

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### Purpose

To compare the measured voltage of non-standard vinegar cells using copper as the inert cathode with the theoretical cell potential and compare measured pH to calculated values through the Nernst equation.

### Background

The experimental cell uses an inert copper strip as the cathode surface, while the actual reduction is hydrogen ion reduction. The anode is either zinc, aluminum or tin, which will undergo oxidation.

Because vinegar is a weak acetic acid solution, the actual hydrogen ion concentration is much lower than 1.0 M. This means the measured voltage will be lower than the standard SHE-based values, and the Nernst equation can be used to explain the difference.

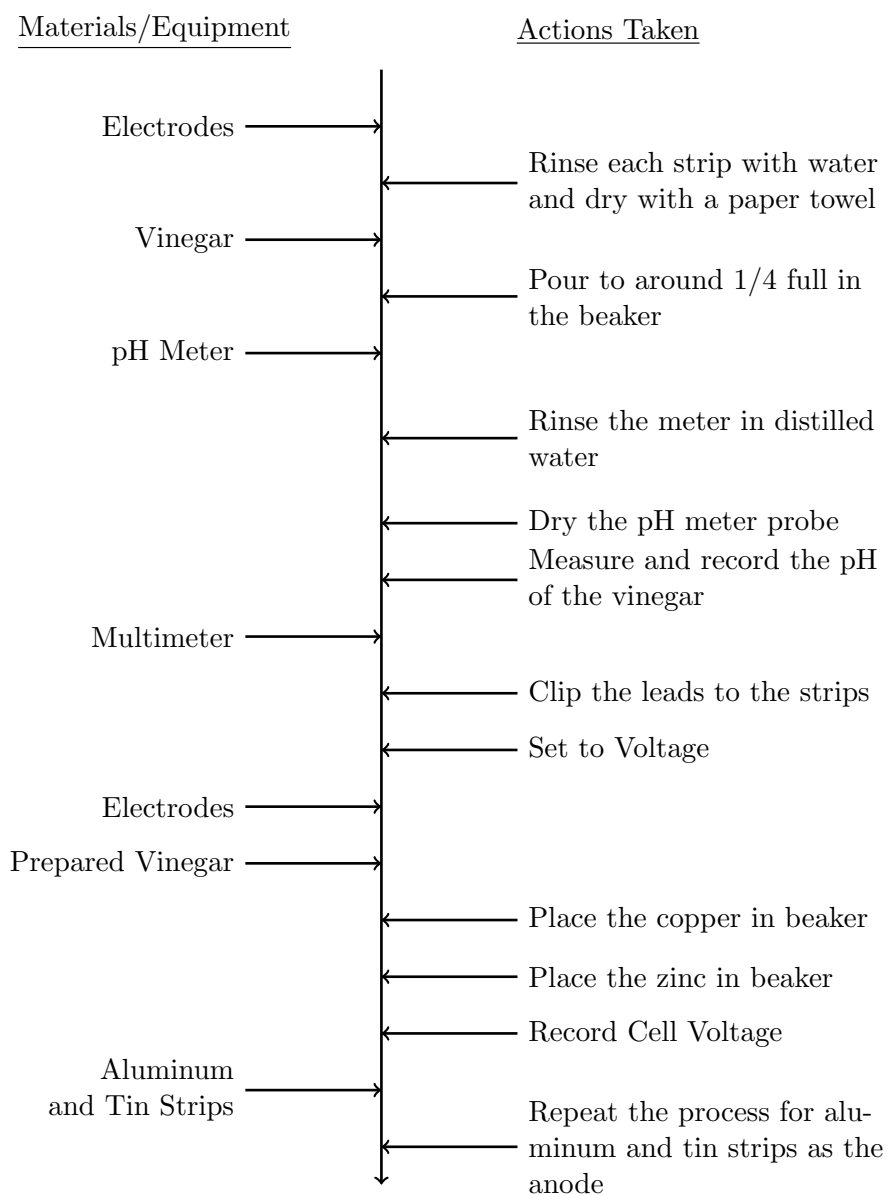
The Nernst equation is written as:

$$E_{cell} = E_{cell}^{\circ} - \frac{0.0592 \text{ V}}{n} \log(Q)$$

### Materials

- 100 mL beaker
- Copper strip (electrode)
- Zinc strip (electrode)
- Aluminum strip (electrode)
- Tin strip (electrode)
- White vinegar (acetic acid solution)
- Multimeter with leads
- Alligator clips
- Distilled water (for rinsing)
- Sandpaper or steel wool (for cleaning metal strips)
- Safety goggles and gloves

## Procedure Flowchart



## Observation

Table 1: Voltage and Qualitative Measurements

| Cell Combination  | Voltage (V) | Observations                             |
|-------------------|-------------|------------------------------------------|
| Copper / Zinc     | 0.94        | Looses a layer?<br>Looks cleaner/shinier |
| Copper / Aluminum | 1.52        |                                          |
| Copper / Tin      | 0.39        |                                          |

Table 2: pH of Vinegar

| Vinegar Sample | pH  |
|----------------|-----|
| Vinegar        | 2.4 |

## Analysis

### Step 1: Prediction of Cell Voltages

To predict the cell potentials ( $E_{\text{cell}}$ ) under non-standard conditions, where the concentration of Hydrogen ions is not 1 M, the Nernst equation may be used.

The Nernst equation at 25°C is:

$$E_{\text{cell}} = E_{\text{cell}}^{\circ} - \frac{0.0592}{2} \log(Q)$$

where  $Q$  is the reaction quotient, which can be expressed as:

$$Q = \frac{[\text{M}^{2+}]}{[\text{H}^+]^2}$$

for Zinc and Tin and

$$Q = \frac{[\text{M}^{3+}]}{[\text{H}^+]^3}$$

for Aluminum, where  $M$  represents the metal undergoing oxidation.

### Find concentration of Hydrogen

Given the measured pH of 2.4, the reactant concentration is:

$$[\text{H}^+] = 10^{-\text{pH}} = 10^{-2.4} = 3.98 \times 10^{-3} \text{ M}$$

### Sample Calculation: Copper-Zinc Cell

Using  $E_{\text{cell}}^{\circ} = +0.76 \text{ V}$  and assuming an initial metal ion concentration of  $[\text{Zn}^{2+}] = 1.0 \times 10^{-5} \text{ M}$ :

$$\begin{aligned} E_{\text{cell}} &= 0.76 - 0.0296 \log\left(\frac{1.0 \times 10^{-5}}{(3.98 \times 10^{-3})^2}\right) \\ &= 0.76 - 0.0296 \log(0.6310) \\ &= 0.76 - 0.0296(-0.200) \\ &= 0.76 + 0.0059 \\ &\approx +0.773615 \text{ V} \\ &\approx +\mathbf{0.77 \text{ V}} \end{aligned}$$

### Step 2: Prediction of pH from Measured Voltage

Using the measured cell voltage from the experimental trials, the Nernst equation can be rearranged to calculate the pH of the electrolyte solution.

### Mathematical Derivation

Starting with the Nernst equation:

$$E_{\text{cell}} = E_{\text{cell}}^{\circ} - 0.0296 \log[\text{M}^{2+}] + 0.0592 \log[\text{H}^+]$$

Sub  $\text{pH} = -\log[\text{H}^+]$ :

$$E_{\text{cell}} = E_{\text{cell}}^{\circ} - 0.0296 \log[\text{M}^{2+}] - 0.0592 \text{ pH}$$

$$E_{\text{cell}} - E_{\text{cell}}^{\circ} + 0.0296 \log[\text{M}^{2+}] = -0.0592 \text{ pH}$$

$$\text{pH} = \frac{E_{\text{cell}}^{\circ} - E_{\text{cell}} - 0.0296 \log[\text{M}^{2+}]}{0.0592}$$

### Sample Calculation: Copper-Zinc System

Using the experimental value  $E_{\text{cell}} = 0.94 \text{ V}$  and assuming a trace metal concentration of  $[\text{Zn}^{2+}] = 1.0 \times 10^{-10} \text{ M}$ :

$$\begin{aligned} \text{pH} &= \frac{0.76 - 0.94 - 0.0296 \log(1.0 \times 10^{-10})}{0.0592} \\ &= \frac{0.76 - 0.94 - 0.0296(-10)}{0.0592} \\ &= \frac{0.76 - 0.94 + 0.296}{0.0592} \\ &= \frac{0.116}{0.0592} \\ &\approx \mathbf{1.96} \end{aligned}$$

## Notes

Qualitative observations of the copper strip in Table1, suggest that a layer of material was removed from the surface, which suggests impurities or contaminants in the experiment.

Table 3: Calculated vs. Measured Values

| Cell System       | Theoretical $E_{\text{cell}}^{\circ}$ (V) | Predicted $E_{\text{cell}}$ (V) | Measured $E_{\text{cell}}$ (V) | Calculated pH from $E_{\text{cell}}$ |
|-------------------|-------------------------------------------|---------------------------------|--------------------------------|--------------------------------------|
| Copper / Zinc     | +0.76                                     | +0.77                           | 0.94                           | 1.96                                 |
| Copper / Aluminum | +1.66                                     | +1.68                           | 1.52                           | 8.06                                 |
| Copper / Tin      | +0.14                                     | +0.15                           | 0.39                           | -8.45                                |

## Discussion

1. One significant source of error in the lab was the presence of impurities on the metal strips, as suggested by the qualitative observations of the copper strip. This could have affected the measured voltage by introducing additional redox reactions or altering the effective surface area of the electrode, leading to higher or lower voltages than expected. Additionally, during the lab, the Vinegar was not replaced between trials, which could have led to changes in the concentration of acetic acid and thus the pH, affecting the voltage measurements for subsequent trials. Furthermore, the metal strips were not cleaned adequately, although washed, they were not sanded, leaving an oxide layer potentially changing results.

2. If there was an opportunity to re-do the lab, I would ensure that the metal strips were thoroughly cleaned and sanded, to eliminate impurities and oxides on the electrodes. Additionally, I would have used fresh vinegar and measured pH each trail. These two changes would eliminate the two largest sources of error.

3. I wanted to make a battery for my DIY lab as I found the topic interesting. Researching what electrolyte to use I remembered the conditions for  $\delta E^{\circ}$  had to be under 1M conditions and wondered if the variation in voltage from  $\delta E^{\circ}$  could be used to calculate pH. A quick search returned the Nernst equation and I chose the lab there.